COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	16/10/2026	**DATE SUBMITTED:**	16/10/2026
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Hands-on Activity 13.1 Parallel Algorithms and Multithreading

1. PROCEDURE OUTPUT

ILO.A

Screenshot	Analysis

CODE:
`#include <iostream>`
`#include <thread>`
`#include <string>`
 	
`void print(int n, const std::string &str) {`
`  std::cout << "Printing integer: " << n << std::endl;`
`  std::cout << "Printing string: " << str << std::endl;`
`}`
 	
`int main() {`
`  std::thread t1(print, 10, "T.I.P.");`
`  t1.join();`
`  return 0;`
`}`
OUTPUT:
`Printing integer: 10`
`Printing string: T.I.P.`
	The following code demonstrates the basic demonstration of multithreading as shown by t1 starting as another thread of execution in order to successfully call the print function and the pass arguments TIP and 10 also showing the t1.joint that acts as a note for the compiler to wait for the previous thread to finish so that it will run correctly.

Table 13-1. Simple One-Threaded Example

ILO.B

Screenshot	Analysis

CODE:
main.cpp > main()
`#include <iostream>`
`#include <thread>`
`#include <vector>`
`#include <string>`
 	
`void print(int n, const std::string &str) {`
`  std::string msg = std::to_string(n) + ":" + str;`
`  std::cout << msg << std::endl;`
`}`
 	
`int main() {`
`  std::vector<std::string> s = {`
`  "T.I.P.",`
`  "Competent",`
`  "Computer",`
`  "Engineers"`
`  };`
	The following code similar to the previous one we've done also demonstrates multithreading that to print the following string TIP, competent, computer, and engineers in the main drive also shows a loop that creates 4 different threads but while the logic remains the same the previous things mentioned were the distinct differences between how the code runs compared to the first one

```
18
19      std::vector<std::thread> threads;
20
21      for (int i = 0; i < s.size(); i++) {
22          threads.push_back(std::thread(print, i, s[i]));
23      }
24
25      for (auto &th : threads) {
26          th.join();
27      }
28
29      return 0;
30  }
```

OUTPUT::

```
PS C:\Users\TIPQC\Deskto
0:T.I.P.
1:Competent
3:Engineers
2:Computer
```

Table 13-2. Multithreaded Example.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.
asymthotic analysis

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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